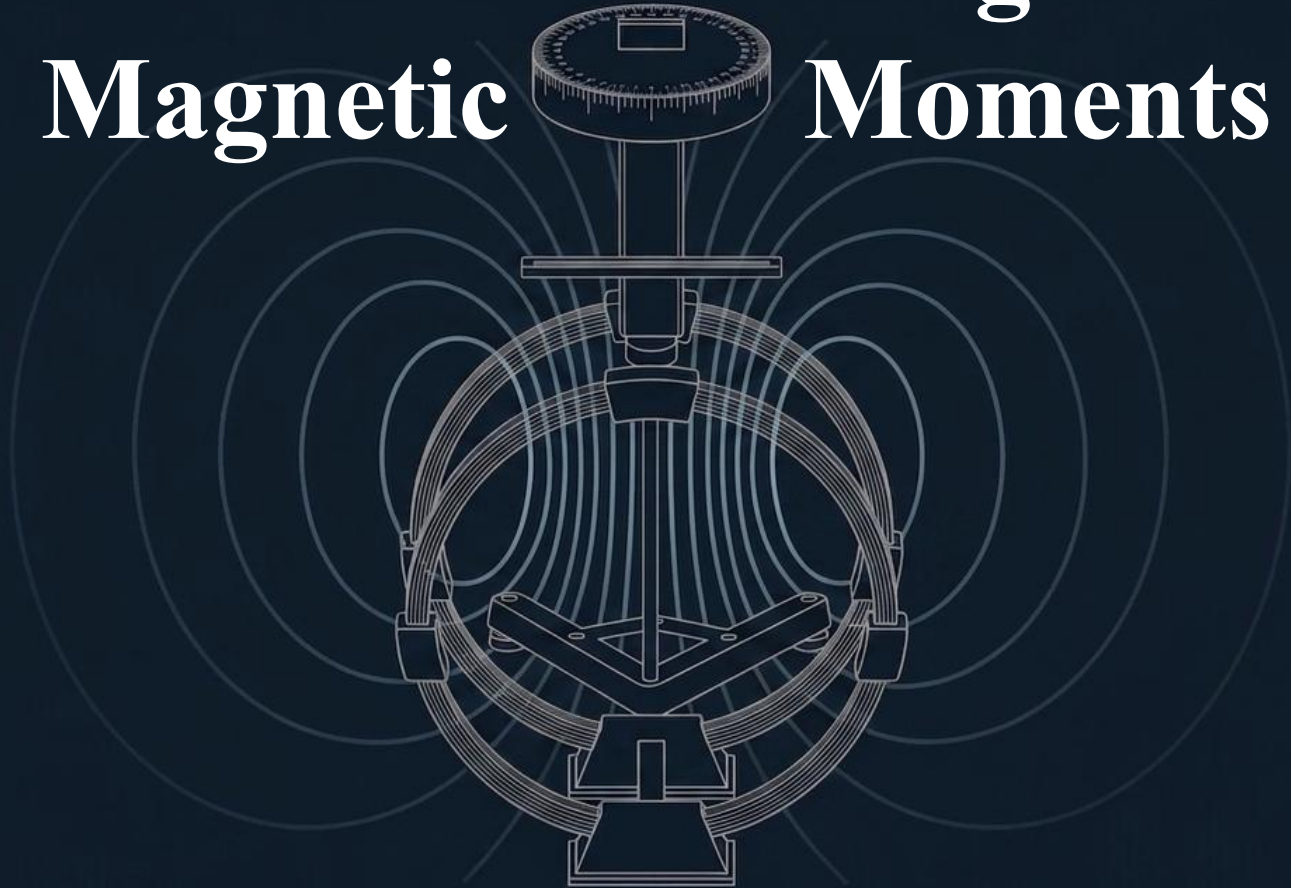


Understanding Magnetic Moments



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Introduction: Motivation and Key Objectives

Motivation

- A current-carrying loop in a magnetic field experiences a torque. This principle forms the basis of electric motors.
- Applications: DC motors, loudspeakers, galvanometers, relays, MRI scanners

Objectives

- To verify the linear relationship between the torque and various independent variables such as the coil current (I_H), number of turns(n), angle(α), sample current (I_L) and sample diameter(d).
- Compare the discrepancy between theoretically calculated Helmholtz constant (c) and experimentally calculated c .
- Use experimentally calculated c to find the area of a star-shaped loop.



Cross-section of a motor working on this principle

Introduction: Underlying Physics

From Lorentz Force to Torque:

1. Lorentz force on a moving charge:

$$\mathbf{F} = q \mathbf{v} \times \mathbf{B}$$

2. For a current element $I d\ell$:

$$d\mathbf{F} = I d\ell \times \mathbf{B}$$

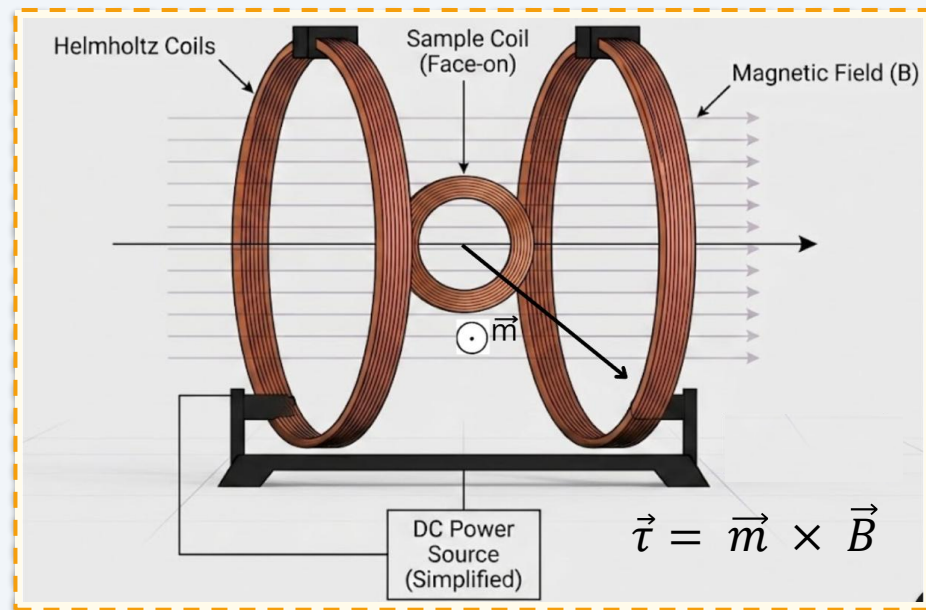
However, for closed loops, the net force is 0 and thus, only a torque acts.

3. Torque about centre:

$$d\mathbf{\tau} = \mathbf{r} \times d\mathbf{F} = \mathbf{r} \times (I d\ell \times \mathbf{B})$$

4. For a planar loop:

$$\mathbf{T} = I \mathbf{A} \mathbf{n} \times \mathbf{B} = \mathbf{m} \times \mathbf{B} \quad (1)$$



Schematic showing directions of different vectors involved.

Introduction: Underlying Physics

Key Equations

Magnetic moment

$$|\vec{m}| = I_L \cdot n \cdot \frac{\pi d^2}{4} \quad (2)$$

Helmholtz field

$$B = \frac{8\mu_0 N I_H}{5\sqrt{5}R} = c I_H \quad (3)$$

Where c is the Helmholtz constant

Torque on loop

$$|\vec{T}| = c \cdot I_H \cdot n \cdot |A| \cdot \sin c \quad (4)$$

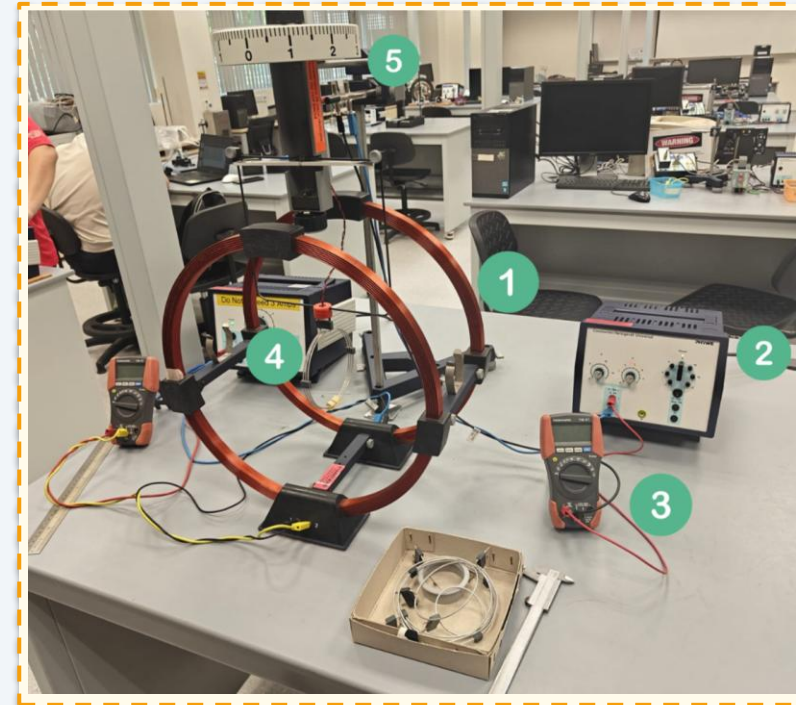
Experimental Setup & Procedure

Core Components:

1. Helmholtz Coils ($N=154$)
2. 2 power sources, one each for sample and coils
3. Ammeters to take current readings
4. Sample holder and sample.
5. Dynamometer to take torque readings

Procedure:

1. Different independent variable for each experiment.
2. Connect sample to the holder and make sure that there is no zero error by adjusting both dials.
3. Set the value of the independent variable (I_H , I_L , n , d , α) and then adjust the top dial.
4. Torque was set and I_H , I_L were adjusted for exp 1 & 4.



Labelled diagram of experimental setup

Results: Theoretical Helmholtz Constant c

Using equation (3) and measured R , we can theoretically calculate the Helmholtz constant c

$$c = \frac{8\mu_0 N}{5\sqrt{5}R}$$

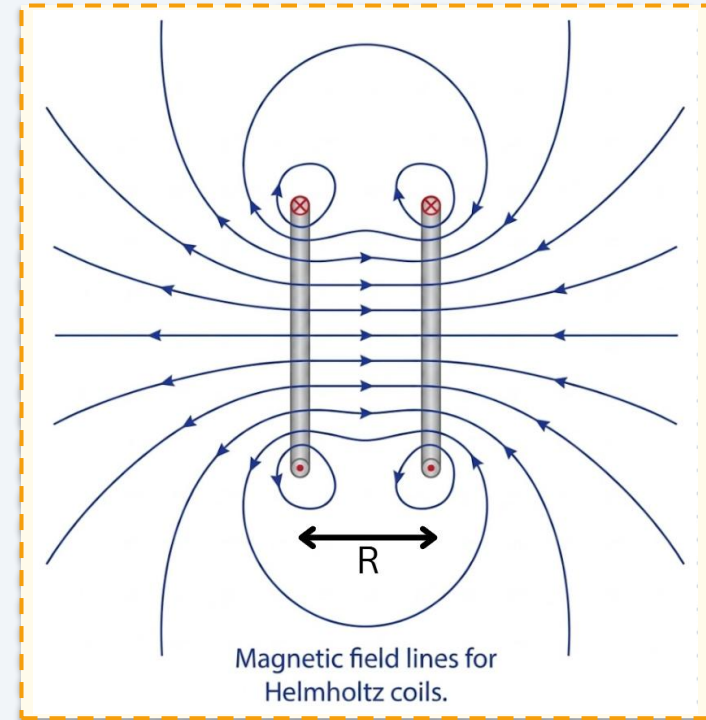
$\mu_0 = 4\pi \times 10^{-7} \text{ T}\cdot\text{m/A}$ (permeability of free space),

$N = 154$ (turns per coil), **$R = 0.1975 \text{ m}$**

Calculated Result:

$$c_{\text{theory}} = 0.000701 \text{ T A}^{-1}$$

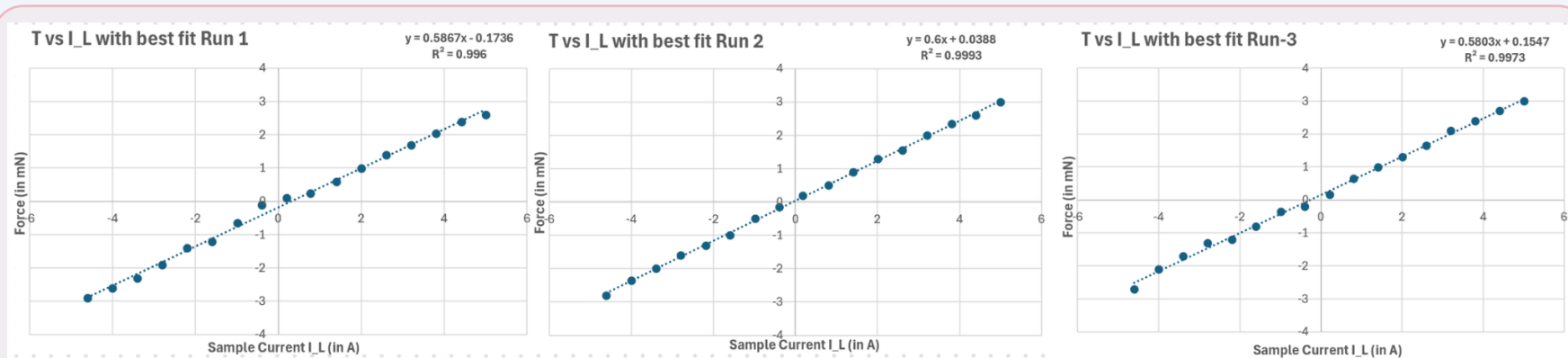
This is the result that will be compared with c_{exp} .



The diagram on the right shows how the arrangement provides a highly uniform $\mathbf{B}$ in the centre.

Results: Raw Data

- Data from each round was plotted immediately after experiment to verify linear trends and detect problems.



Relation between T and I_L using method (3 runs).

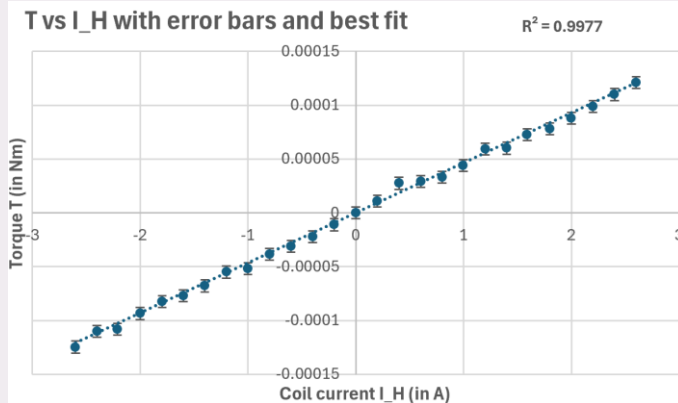
We can see round 1 and 3 have slightly more noise than round 2, and their slopes are close in value but different from round 2 by around 3%.

Results: Error Analysis

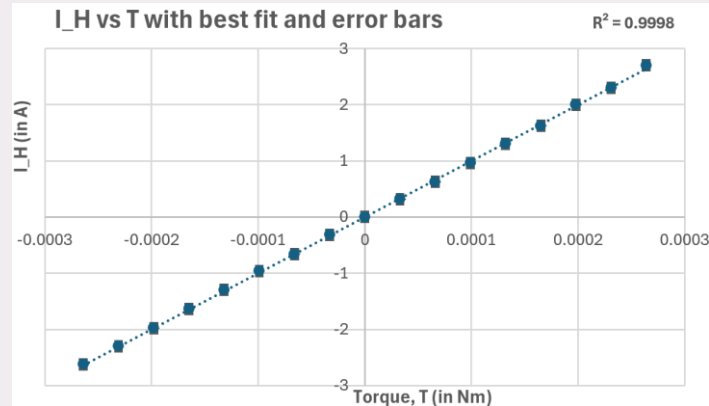
- Error was assumed to be negligible for I_H and I_L when they were kept constant (Exp 2, 3 & 5).
- Error in I_H and I_L was taken to be 0.01 A for exp 1 & 4.
- Error in $\text{angle}(\alpha)$ was assumed to be 1° and considered for all experiments.
- Error in the diameter of sample coils has been taken to be 0.1 mm, although the least count of the vernier callipers is 0.05 mm. This is because some of the circular loops themselves were not uniformly circular, specifically the $n=1$, $d=12\text{cm}$ sample loop.
- Error in Torque has been taken as half the least count of the dynamometer.
- Error in Torque for exp 1 and exp 4 (method 2) has been taken to be 0.5% due to equipment limitations (weight of the connections, friction).

Results: Comparison of Methods

- For exp 1 & 4, we can either set a value of I_H and I_L and then adjust the dynamometer accordingly (method 1), or we can set a value of the torque and then adjust I_H or I_L (method 2).



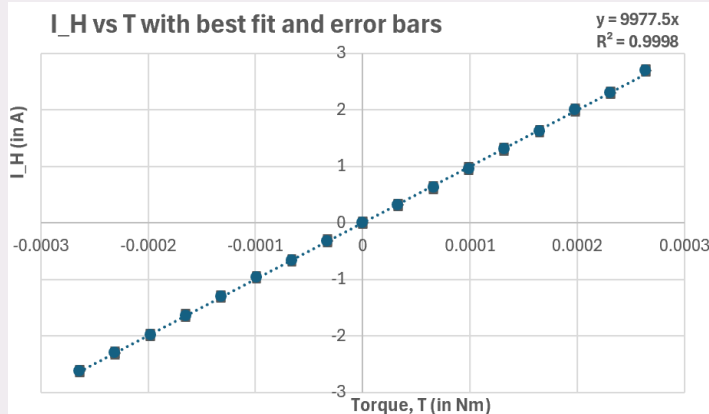
T vs I_H | Method 1
 Percentage uncertainty: 8.74%
 More visible error bars | Lower R^2



I_H vs T | Method 2
 Percentage uncertainty: 0.88%
 Less visible error bars | Higher R^2

- Method 2 removes the drawback of the high least count and gives us a better fit (Higher R^2) and gives us lesser discrepancy with theory.

Results: I_H vs T (Exp 1) & T vs n (Exp 2)

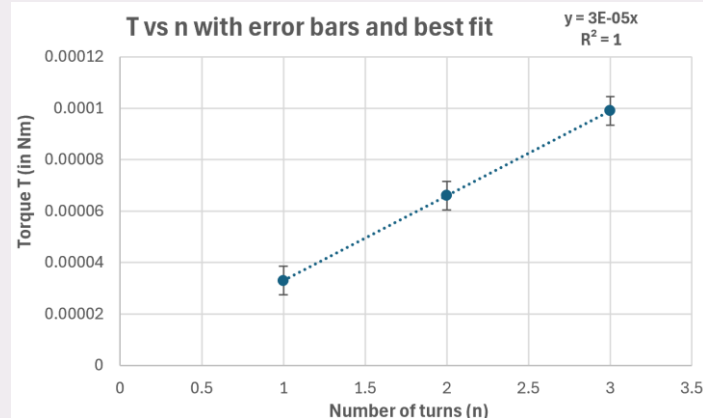


Exp 1: I_H vs T for $n=3$, $I_L=1.8A$, $d=12.00$ cm.
 $c_1 = 4 \times k / (n \times d^2 \times \pi \times I_L)$

Exp. Slope
 $k = 9977.47 \text{ A(Nm)}^{-1}$

Exp. Helmholtz Constant
 $C_1 = (7.4 \pm 0.0653) \times 10^{-4} \text{ T A}^{-1}$

Discrepancy from Theoretical value: 5.54%



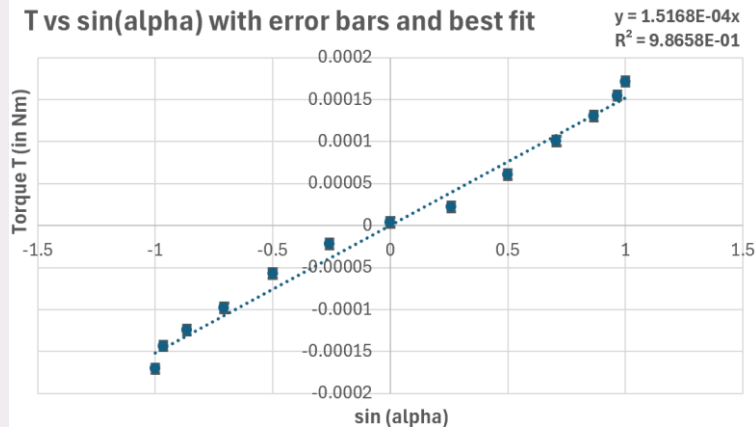
Exp 2: T vs n for $I_H=2.6$, $I_L=1.5A$, $d=12.00$ cm.
 $c_2 = 4 \times k / (I_H \times d^2 \times \pi \times I_L)$

Exp. Slope
 $k = 0.000033 \text{ Nm}$

Exp. Helmholtz Constant
 $c_2 = (7.5 \pm 0.626) \times 10^{-4} \text{ T A}^{-1}$

Discrepancy from Theoretical value: 6.51%

Results: T vs $\sin\alpha$ (Exp 3) & I_L vs T (Exp 4)



Exp 3: T vs $\sin\alpha$: $n=3$, $I_L=1.8\text{A}$, $I_H=2.6\text{A}$, $d=12.00\text{cm}$.

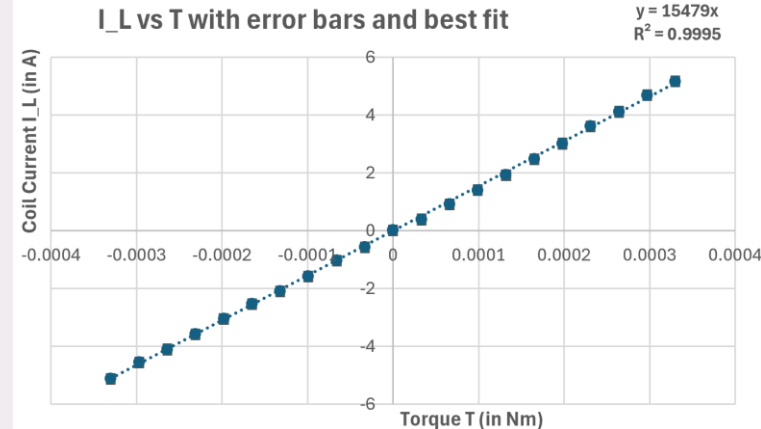
Exp. Slope

$$k = 0.000152 \text{ Nm}$$

Exp. Helmholtz Constant

$$c_3 = (6.9 \pm 0.397) \times 10^{-4} \text{ T A}^{-1}$$

Discrepancy from Theoretical value: 1.59%



Exp 4: I_L vs T : $I_H=2.6$, $n=3$, $d=12.00\text{cm}$.

Exp. Slope

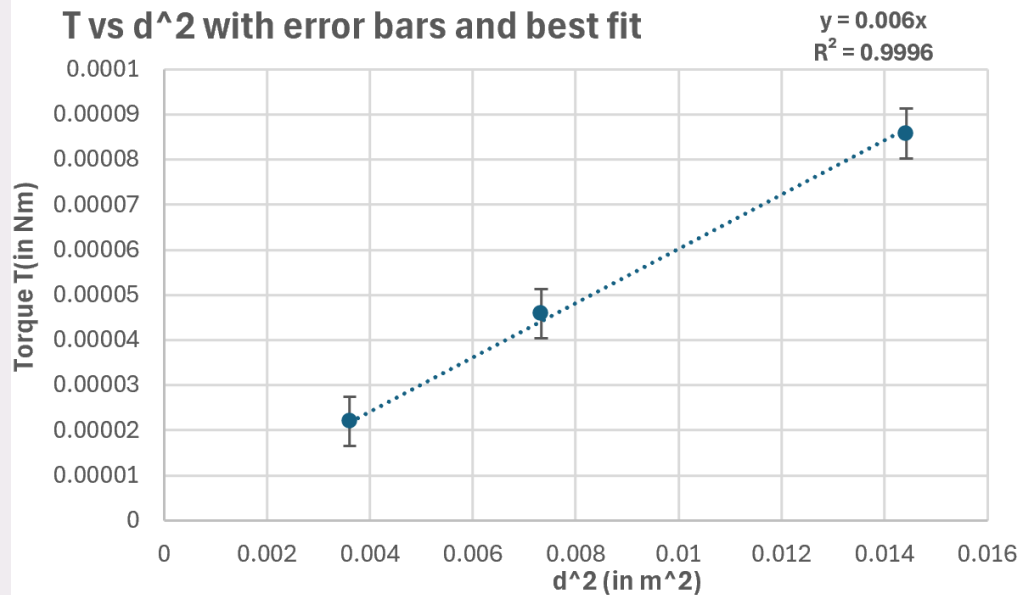
$$k = 15479.08 \text{ A T}^{-1}$$

Exp. Helmholtz Constant

$$c_4 = (7.3 \pm 0.047) \times 10^{-4} \text{ T A}^{-1}$$

Discrepancy from Theoretical value: 4.12%

Results: Torque vs. Loop Diameter (Exp 5)



T vs d^2 for $I_H=2.6$, $I_L = 4$ A, $n=1$
 $c_5 = 4k/(I_H \times I_L \times n \times \pi)$

Value of slope

$k = 0.00587 \text{ Nm}^{-1}$

Exp. Helmholtz Constant

$c_5 = (7.2 \pm 0.753) \times 10^{-4} \text{ T A}^{-1}$

Discrepancy from Theory

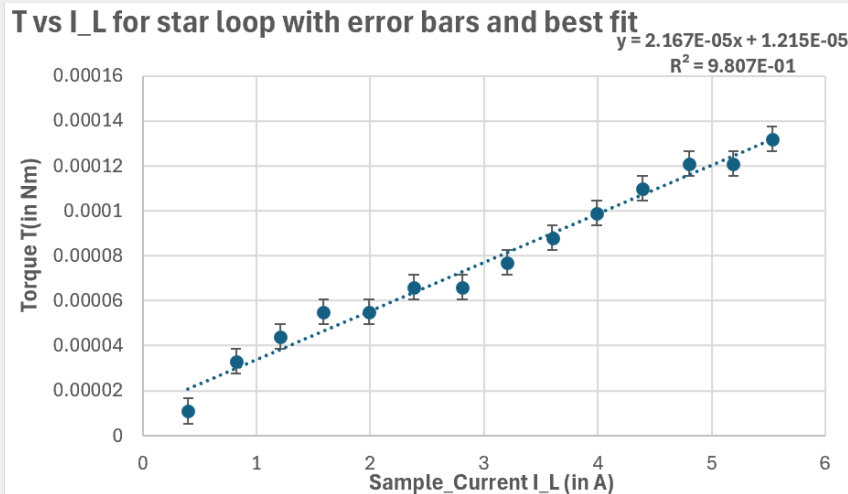
2.69%

- X-axis: Diameter d^2 (m^2)
- Relationship: Quadratic ($T \propto d^2$)
- Loops: 6.01, 8.56, 12.00 cm

Results: Determination of Star Area (Exp 6)

- The theoretical area is determined by approximating the star as a combination of a regular pentagon and 5 isosceles triangles.

$$A_{\text{theoretical}} = 0.009967 \text{ m}^2$$



T vs n for $I_H = 2.6$, star loop

$$A_{\text{exp}} = k / (I_H \times c_4)$$

Value of Slope

$$k = 0.0000216 \text{ T A}^{-1}$$

Experimental Value of A

$$A = 110 \pm 8.128 \text{ cm}^2$$

Discrepancy from Theory

9.4%

- X-axis:** Sample current (I_L)
- Relationship:** Linear ($T \propto I_L$)

Discussion: Summary of Results

Exp No.	Variable	Value ($\times 10^{-4}$ for 1-5)	Units	Discrepancy	Deviation from c_{mean}
1	c_1	7.4 ± 0.0653	T A^{-1}	5.54%	+1.93%
2	c_2	7.5 ± 0.626	T A^{-1}	6.51%	+3.31%
3	c_3	6.9 ± 0.397	T A^{-1}	1.59%	-4.96%
4	c_4	7.3 ± 0.047	T A^{-1}	4.12%	+0.55%
5	c_5	7.2 ± 0.753	T A^{-1}	2.69%	-0.83%
6	Star Area	110 ± 8.128	cm^2	9.4%	N/A

Choice of final experimental value of c

c_4 was chosen as it has significantly lower uncertainty compared to values from exp 2, 3 & 5. It shows lower discrepancy from the theoretical value than c_1 .

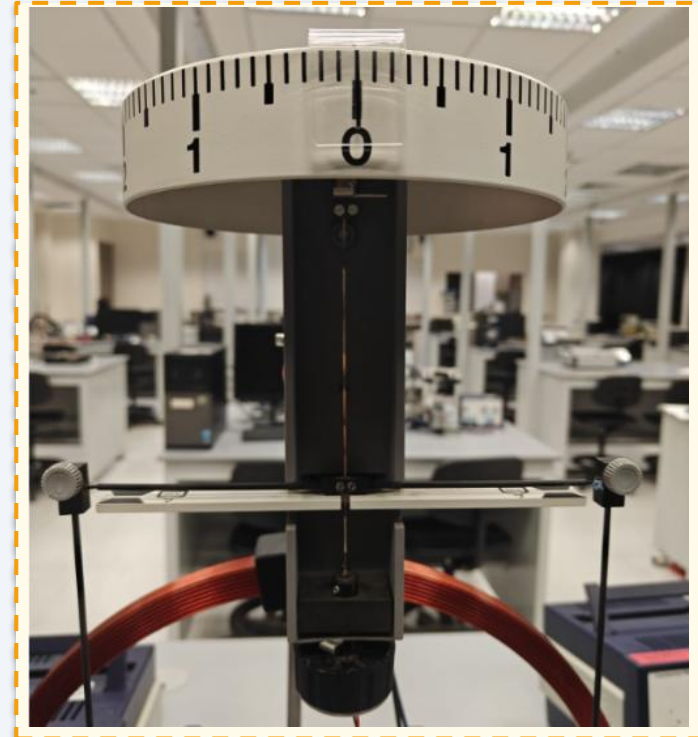
$$c_{\text{exp}} = 7.3 \pm 0.047 \text{ T A}^{-1}$$

Discussion: Dominant Sources of Error

- The largest contributor of uncertainty is the limited resolution of the dynamometer for exp 2,3 & 5.
- For exp 1 & 4, the error in I_H/I_L and error in T contribute almost equally, with error in T contributing slightly more.
- Discrepancy in the star loop area value is due to systemic factors such as the overlapping of wire and the geometry of the star loop.

Additional possible systemic errors:

- Orientation and placement of the sample.
- Geometry of the sample.
- Friction at the contacts and between the dynamometer levers and the platforms.
- Vibrations introduced while turning dynamometer dials.



Limited resolution and multiple friction points of the torsion dynamometer are visible.

Conclusion

- Verified $\mathbf{T} = \mathbf{m} \times \mathbf{B}$ across all five independent variables
- Linear dependences confirmed for I_H , I_L , $\sin \alpha$ and n . Quadratic dependence on d .
- Discrepancy between magnetic area of star loop and physical area of star point towards systemic factors.

Key Results

$$c_{\text{exp}} = (7.3 \pm 0.047) \times 10^{-4} \text{ T A}^{-1}$$

$$\text{Discrepancy} = 4.12\%$$

$$\text{Star Loop Area} = (110 \pm 8.128) \text{ cm}^2$$

$$\text{Discrepancy} = 9.4\%$$

Possible Improvements

- Dynamometer with higher resolution (lower least count).
- A more precise way to measure angles.
- Introducing coil samples with $n > 3$ and intermediate d values between 6cm and 12cm.

Thank You!

Open for Questions

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